

University of Minnesota
School of Physics and Astronomy

2026 Spring Physics 8012
Quantum Field Theory II
Assignment Solution

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Homework 1 Due to January 29th 11:00 AM

Question 1

The constant J in the Heisenberg model (slide 6) can be positive or negative. The model can describe either ferromagnetism or anti-ferromagnetism. To which of these two cases positive/ negative J corresponds? Why?

$$\mathcal{H} = - \sum_{\langle ij \rangle} J \mathbf{S}_i \cdot \mathbf{S}_j \quad (1)$$

Answer

Since **the Hamiltonian is minimized when the energy is the lowest**, we can analyze the two cases as follows:

- For $J > 0$, the Hamiltonian becomes $\mathcal{H} = - \sum_{\langle ij \rangle} |J| \mathbf{S}_i \cdot \mathbf{S}_j$. To minimize the energy, the spins $\mathbf{S}_i$ and $\mathbf{S}_j$ should align parallel to each other, leading to ferromagnetism.
- For $J < 0$, the Hamiltonian becomes $\mathcal{H} = - \sum_{\langle ij \rangle} -|J| \mathbf{S}_i \cdot \mathbf{S}_j = \sum_{\langle ij \rangle} |J| \mathbf{S}_i \cdot \mathbf{S}_j$. To minimize the energy, the spins $\mathbf{S}_i$ and $\mathbf{S}_j$ should align anti-parallel to each other, leading to anti-ferromagnetism.

□

Question 2

Pirates found a parchment describing location of treasure trove hidden somewhere on the island by their predecessors . The note reads:

Start from the capsized boat. Go in the direction of the palm tree carefully counting the number of steps. When you reach the palm tree turn exactly right and make exactly the same number of steps. Mark the point you arrived at. Then return to the boat. Go in the direction of the rock counting the number of steps. When you reach the rock turn left. After having made the same number of steps mark the second point. The treasure trove is in the middle of the line connecting two marked points. The problem is that the capsized boat was nowhere in sight, it had disappeared. How can the pirates still find the treasure trove?

Answer

First let's denote the position of the palm tree as point $P(x_1, y_1)$ and the position of the rock as point $R(x_2, y_2)$. We also denote the unknown position of the capsized boat as point $B(x, y)$.

When the pirates walk from the boat to the palm tree, they cover a distance equal to the length of the vector $\overrightarrow{BP} = (x_1 - x, y_1 - y)$. After reaching the palm tree, they turn right and walk the same distance, which corresponds to moving in the direction perpendicular to $\overrightarrow{BP}$. Let's say the right turn results in a new vector $\overrightarrow{PR} = (y_1 - y, -(x_1 - x))$. The coordinates of the first marked point M_1 can be expressed as:

$$M_1 = P + \overrightarrow{PR} = (x_1 + (y_1 - y), y_1 - (x_1 - x)) = (x_1 + y_1 - y, y_1 - x_1 + x) \quad (2)$$

Similarly, when the pirates walk from the boat to the rock, they cover a distance equal to the length of the vector $\overrightarrow{BR} = (x_2 - x, y_2 - y)$. After reaching the rock, they turn left and walk the same distance, which corresponds to moving in the direction perpendicular to $\overrightarrow{BR}$. Let's say the left turn results in a new vector $\overrightarrow{RL} = (-(y_2 - y), x_2 - x)$. The coordinates of the second marked point M_2 can be expressed as:

$$M_2 = R + \overrightarrow{RL} = (x_2 - (y_2 - y), y_2 + (x_2 - x)) = (x_2 - y_2 + y, y_2 + x_2 - x) \quad (3)$$

The treasure trove is located at the midpoint of the line segment connecting the two marked points M_1 and M_2 . The coordinates of the midpoint T can be calculated as:

$$T = \left(\frac{(x_1 + y_1 - y) + (x_2 - y_2 + y)}{2}, \frac{(y_1 - x_1 + x) + (y_2 + x_2 - x)}{2} \right) \quad (4)$$

Simplifying the expressions for the coordinates of T , we get:

$$T = \left(\frac{x_1 + x_2 + y_1 - y_2}{2}, \frac{y_1 + y_2 - x_1 + x_2}{2} \right) \quad (5)$$

Notice that the coordinates of the treasure trove T do not depend on the unknown position of the boat $B(x, y)$. Therefore, the pirates can find the treasure trove using only the known positions of the palm tree

and the rock, without needing to know the location of the capsized boat.

□

Homework 2 Due to February 5 9:00 AM

Question 1

Consider a set of three differential operators

$$\left\{ T^+ = x^2 \frac{d}{dx}, \quad T^0 = -x \frac{d}{dx}, \quad T^- = \frac{d}{dx} \right\} \quad (6)$$

acting on an arbitrary smooth function $f(x)$. Find the algebra of these operators, i.e. all three commutators $[T^i, T^j]$ where $i, j = +, -, 0$. Show that this algebra is closed.

Answer

- $[T^i, T^i]$: $[T^i, T^i] = 0$ for $i = +, -, 0$. It's obvious since any operator commutes with itself.
- $[T^+, T^-]$:

$$\begin{aligned} [T^+, T^-]f(x) &= T^+T^-f(x) - T^-T^+f(x) \\ &= x^2 \frac{d}{dx} \left(\frac{df(x)}{dx} \right) - \frac{d}{dx} \left(x^2 \frac{df(x)}{dx} \right) \\ &= x^2 \frac{d^2 f(x)}{dx^2} - \left(2x \frac{df(x)}{dx} + x^2 \frac{d^2 f(x)}{dx^2} \right) \\ &= -2x \frac{df(x)}{dx} \\ &= 2T^0 f(x) \end{aligned} \quad (7)$$

- $[T^0, T^-]$:

$$\begin{aligned} [T^0, T^-]f(x) &= T^0T^-f(x) - T^-T^0f(x) \\ &= -x \frac{d}{dx} \left(\frac{df(x)}{dx} \right) - \frac{d}{dx} \left(-x \frac{df(x)}{dx} \right) \\ &= -x \frac{d^2 f(x)}{dx^2} + \left(\frac{df(x)}{dx} + x \frac{d^2 f(x)}{dx^2} \right) \\ &= \frac{df(x)}{dx} \\ &= -T^- f(x) \end{aligned} \quad (8)$$

- $[T^+, T^0]$:

$$\begin{aligned}
[T^+, T^0]f(x) &= T^+T^0f(x) - T^0T^+f(x) \\
&= x^2 \frac{d}{dx} \left(-x \frac{df(x)}{dx} \right) - \left(-x \frac{d}{dx} \left(x^2 \frac{df(x)}{dx} \right) \right) \\
&= -x^3 \frac{d^2f(x)}{dx^2} - x^2 \frac{df(x)}{dx} + \left(2x^2 \frac{df(x)}{dx} + x^3 \frac{d^2f(x)}{dx^2} \right) \\
&= x^2 \frac{df(x)}{dx} \\
&= T^+f(x)
\end{aligned} \tag{9}$$

Hence, the algebra of these operators is

$$\begin{aligned}
[T^+, T^-] &= 2T^0, \\
[T^-, T^0] &= T^-, \\
[T^+, T^0] &= T^+.
\end{aligned} \tag{10}$$

This algebra is closed since the commutators of any two operators in the set can be expressed as a linear combination of the operators in the set. $\square$

Question 2

Compare it with the algebra of three Pauli matrices:

$$\left\{ \frac{1}{2}\sigma_1, \quad \frac{1}{2}\sigma_2, \quad \frac{1}{2}\sigma_3 \right\} \quad (11)$$

How this algebra is called? Find the linear combinations of the Pauli matrices in this which form exactly the same algebra as the operators in question 1. What is the difference between the representations of the two algebras above?

Answer

The algebra of the three Pauli matrices is given by the commutation relations:

$$\left[\frac{1}{2}\sigma_i, \frac{1}{2}\sigma_j \right] = i\epsilon_{ijk} \frac{1}{2}\sigma_k, \quad (12)$$

where ϵ_{ijk} is the Levi-Civita symbol. This algebra is called the $\mathfrak{su}(2)$ algebra, which is the Lie algebra of the group $SU(2)$.

Next, we consider the following linear combinations of the Pauli matrices:

$$\begin{aligned} J^+ &= \frac{1}{2}(\sigma_1 + i\sigma_2), \\ J^- &= \frac{1}{2}(\sigma_1 - i\sigma_2), \\ J^0 &= \frac{1}{2}\sigma_3. \end{aligned} \quad (13)$$

Then we can compute their commutation relations:

$$\begin{aligned} [J^+, J^-] &= \frac{1}{4}[\sigma_1 + i\sigma_2, \sigma_1 - i\sigma_2] \\ &= \frac{1}{4}([\sigma_1, \sigma_1] - i[\sigma_1, \sigma_2] + i[\sigma_2, \sigma_1] + [\sigma_2, \sigma_2]) \\ &= \frac{1}{4}(0 - i(2i\sigma_3) + i(-2i\sigma_3) + 0) \\ &= \sigma_3 \\ &= 2J^0, \end{aligned} \quad (14)$$

$$\begin{aligned}
[J^0, J^-] &= \frac{1}{4}[\sigma_3, \sigma_1 - i\sigma_2] \\
&= \frac{1}{4}([\sigma_3, \sigma_1] - i[\sigma_3, \sigma_2]) \\
&= \frac{1}{4}(2i\sigma_2 - i(-2i\sigma_1)) \\
&= -\frac{1}{2}(\sigma_1 - i\sigma_2) \\
&= -J^-,
\end{aligned} \tag{15}$$

$$\begin{aligned}
[J^+, J^0] &= \frac{1}{4}[\sigma_1 + i\sigma_2, \sigma_3] \\
&= \frac{1}{4}([\sigma_1, \sigma_3] + i[\sigma_2, \sigma_3]) \\
&= \frac{1}{4}(-2i\sigma_2 + i(2i\sigma_1)) \\
&= \frac{1}{2}(\sigma_1 + i\sigma_2) \\
&= J^+.
\end{aligned} \tag{16}$$

Thus, the commutation relations of J^+ , J^- , and J^0 are:

$$\begin{aligned}
[J^+, J^-] &= 2J^0, \\
[J^-, J^0] &= J^-, \\
[J^+, J^0] &= J^+.
\end{aligned} \tag{17}$$

These commutation relations are exactly the same as those of the operators T^+ , T^- , and T^0 from question 1.

Next, the essential difference between the two representations of the algebras is that the operators T^+ , T^- , and T^0 act on an infinite-dimensional space of smooth functions. However, the Pauli matrices J^+ , J^- , and J^0 act on a two-dimensional complex vector space (spinor space). Therefore, while the algebraic structures are identical, their representations differ significantly in terms of the spaces on which they act. $\square$

Homework 3 Due to February 12 8:00 AM

Question 1

- (a) Write down the generators of the SU(3) group in the fundamental representation. Use the standard form of the Gell-Mann matrices. Commute them and find the set of the structure constants for SU(3).
- (b) Calculate anti-commutators of the same generators. The constants in the right-hand-side of the anti-commutators are called d symbols.
- (c) Compare the result to anti-commutators for SU(2) (in which the generators in fundamental representation are $(1/2) \times$ Pauli matrices). What is the qualitative difference?
- (d) For $N = 3$ check the the following equation is valid:

$$f^{abc} f^{adg} = \frac{2}{N} (\delta^{bd} \delta^{cg} - \delta^{bg} \delta^{cd}) + d^{abd} d^{acg} - d^{acd} d^{abg} \quad (18)$$

Answer

(a)

The generators of the SU(3) group in the fundamental representation are given by the Gell-Mann matrices divided by 2:

$$T^1 = \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad T^2 = \frac{1}{2} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad T^3 = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (19)$$

$$T^4 = \frac{1}{2} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad T^5 = \frac{1}{2} \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \quad T^6 = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad (20)$$

$$T^7 = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad T^8 = \frac{1}{2\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}. \quad (21)$$

The commutation relations for the generators can be expressed in terms of the structure constants f^{abc} as follows:

$$f^{abc} = \frac{2}{i} \text{Tr}(T^a [T^b, T^c]) \quad (22)$$

Calculating the commutators and using the above formula, we find the non-zero structure constants for SU(3)

(see the calculation details in the Mathematica notebook):

$$f^{123} = 1, \quad f^{147} = \frac{1}{2}, \quad f^{156} = -\frac{1}{2}, \quad f^{246} = \frac{1}{2}, \quad f^{257} = \frac{1}{2}, \quad f^{345} = \frac{1}{2}, \quad f^{367} = -\frac{1}{2}, \quad (23)$$

$$f^{458} = \frac{\sqrt{3}}{2}, \quad f^{678} = \frac{\sqrt{3}}{2}. \quad (24)$$

(b)

The anti-commutators of the generators can be expressed in terms of the d symbols as follows:

$$d^{abc} = 2\text{Tr}(T^a\{T^b, T^c\}) \quad (25)$$

Calculating the anti-commutators and using the above formula, we find the non-zero d symbols for $\text{SU}(3)$ (see the calculation details in the Mathematica notebook):

$$d^{118} = d^{228} = d^{338} = -d^{888} = \frac{1}{\sqrt{3}}, \quad d^{448} = d^{558} = d^{668} = d^{778} = -\frac{1}{2\sqrt{3}}, \quad (26)$$

$$d^{146} = d^{157} = -d^{247} = d^{256} = d^{344} = d^{355} = -d^{366} = -d^{377} = \frac{1}{2}. \quad (27)$$

Note that the anti-commutators yield a more complex structure involving the d symbols, which is

$$\{T^a, T^b\} = \frac{1}{3}\delta^{ab}I + d^{abc}T^c. \quad (28)$$

However, we can also derive d^{abc} using the relation:

$$d^{abc} = 2\text{Tr}(T^a\{T^b, T^c\}), \quad (29)$$

since the generators are traceless.

(c)

For $\text{SU}(2)$, the generators in the fundamental representation are given by $(1/2)$ times the Pauli matrices:

$$T^1 = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad T^2 = \frac{1}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad T^3 = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (30)$$

Calculating the anti-commutators for $\text{SU}(2)$, we find that they are proportional to the identity matrix:

$$\{T^a, T^b\} = \frac{1}{2}\delta^{ab}I. \quad (31)$$

The qualitative difference between $\text{SU}(2)$ and $\text{SU}(3)$ is that for $\text{SU}(2)$, the anti-commutators yield a simple result proportional to the identity matrix, while for $\text{SU}(3)$, the anti-commutators yield a more complex structure involving the d symbols, which are not simply proportional to the identity matrix. This reflects the richer structure of the $\text{SU}(3)$ group compared to $\text{SU}(2)$.

(d)

Please refer to the Mathematica notebook for the detailed calculation of the equation, and the result is valid. $\square$

Homework 4 Due to February 19 8:00 AM

Question 1

Assume we have Yang-Mills theory with the gauge sector

$$G = \text{SU}(3) \quad (32)$$

and the Higgs sector consisting of one real scalar field in the *adjoint* representation of $\text{SU}(3)$.

Assuming that the adjoint field develops a vacuum expectation value

$$\langle \phi^a \phi^a \rangle = v^2 \quad (33)$$

(large on the scale of quantum corrections) determine the masses of all eight gauge bosons in terms of the above expectation value v .

Hint: First find the simplest form to which one can reduce an x -independent generic adjoint field (with the constraint (2)) by using $\text{SU}(3)$ rotations. In the matrix form you can write ϕ as $\phi^a T^a$ where T^a are the $\text{SU}(3)$ generators (proportional to the Gell-Mann matrices). How many (independent) real quantities parametrize this generic form?

Answer

We first write down the Lagrangian of the theory:

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu} + \frac{1}{2} (D_\mu \phi)(D^\mu \phi) - V(\phi), \quad (34)$$

where $F_{\mu\nu}^a$ is the field strength tensor for the gauge fields, D_μ is the covariant derivative, ϕ is the adjoint scalar field, and $V(\phi)$ is the potential for the scalar field. To be specific, the ϕ field has 8 components, and can be written as $\phi = \phi^a T^a$, where T^a are the generators of $\text{SU}(3)$. The covariant derivative is given by

$$D_\mu = \partial_\mu - ig A_\mu^a T^a, \quad (35)$$

where g is the gauge coupling constant and A_μ^a are the gauge fields, T^a are the generators of $\text{SU}(3)$ in the adjoint representation. The field strength tensor is given by

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + gf^{abc} A_\mu^b A_\nu^c, \quad (36)$$

where f^{abc} are the structure constants of SU(3). Notice that if we consider $D_\mu\phi$, we can get

$$D_\mu\phi = \partial_\mu\phi - igA_\mu^a T_{\text{adj}}^a \phi \quad (37)$$

$$D_\mu\phi_a = \partial_\mu\phi_a - igA_\mu^b (T_{\text{adj}}^b)_{ac} \phi_c, \quad \text{where } (T_{\text{adj}}^b)_{ac} = -if^{bac} \quad (38)$$

$$= \partial_\mu\phi_a - gA_\mu^b f^{bac} \phi_c \quad (39)$$

$$= \partial_\mu\phi_a + gf^{abc} A_\mu^b \phi_c. \quad (40)$$

We can write down the ϕ field in the matrix form Φ as

$$\Phi = \sum_{a=1}^8 \phi^a T^a, \quad (41)$$

where T^a are the generators of SU(3) in the fundamental representation, which can be expressed in terms of the Gell-Mann matrices λ^a as $T^a = \frac{1}{2}\lambda^a$. The Gell-Mann matrices are traceless and Hermitian, and they form a basis for the Lie algebra of SU(3). Note that $\text{Tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$, which means that the generators are normalized such that the trace of their product is proportional to the Kronecker delta. Now we can rewrite whole Lagrangian in terms of Φ as

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu} + \text{Tr}[(D_\mu\Phi)(D^\mu\Phi)] - V(\Phi), \quad (42)$$

where the covariant derivative is now given by

$$D_\mu\Phi = \partial_\mu\Phi - ig[A_\mu, \Phi], \quad (43)$$

and $V(\Phi)$ is given by

$$V(\Phi) = -\frac{1}{2}\mu^2 \text{Tr}(\Phi^2) + \frac{1}{4}\lambda \text{Tr}(\Phi^4). \quad (44)$$

The vacuum expectation value of Φ can be written as

$$v^2 = 2\langle \text{Tr}(\Phi^2) \rangle = \langle \sum_{a=1}^8 \phi^a \phi^a \rangle \quad (45)$$

However, because Φ is a traceless Hermitian matrix, we can do a SU(3) rotation to Φ to make it diagonal, and the diagonal form of Φ can be written as

$$\Phi = \begin{pmatrix} \phi_1 & 0 & 0 \\ 0 & \phi_2 & 0 \\ 0 & 0 & \phi_3 \end{pmatrix}, \quad (46)$$

where ϕ_1 , ϕ_2 , and ϕ_3 are the eigenvalues of Φ . Since Φ is traceless, we have

$$\phi_1 + \phi_2 + \phi_3 = 0. \quad (47)$$

Thus, the one of the convenient choices of the vacuum expectation value of $\langle \Phi \rangle$ is

$$\langle \Phi \rangle = v_3 T^3 + v_8 T^8 = \begin{pmatrix} \frac{v_3}{2} + \frac{v_8}{2\sqrt{3}} & 0 & 0 \\ 0 & -\frac{v_3}{2} + \frac{v_8}{2\sqrt{3}} & 0 \\ 0 & 0 & -\frac{v_8}{\sqrt{3}} \end{pmatrix}, \quad (48)$$

where v_3 and v_8 are the vacuum expectation values of the components of Φ along the T^3 and T^8 directions, respectively. Note that $v_3^2 + v_8^2 = v^2$. The mass matrix for the gauge bosons can be obtained from the kinetic term of the scalar field, which is given by

$$\text{Tr}[(D_\mu \Phi)(D^\mu \Phi)] = \text{Tr}[(\partial_\mu \Phi - ig[A_\mu, \Phi])(\partial^\mu \Phi - ig[A^\mu, \Phi])] \quad (49)$$

$$= \text{Tr}[(\partial_\mu \Phi)(\partial^\mu \Phi)] - 2ig \text{Tr}[(\partial_\mu \Phi)[A^\mu, \Phi]] - g^2 \text{Tr}[[A_\mu, \Phi][A^\mu, \Phi]]. \quad (50)$$

Then we can expand this term to get the mass terms for the gauge bosons. The third term in the above expression gives the mass terms for the gauge bosons, which can be written as

$$-g^2 \text{Tr}[[A_\mu, \Phi][A^\mu, \Phi]] \quad (51)$$

$$\Rightarrow -g^2 \text{Tr}[[A_\mu, \langle \Phi \rangle][A^\mu, \langle \Phi \rangle]] = -g^2 \text{Tr}[[T^a, v_3 T^3 + v_8 T^8][T^b, v_3 T^3 + v_8 T^8]] A_\mu^a A^{\mu b} \quad (52)$$

$$= -\frac{1}{2} M_{ab} A_\mu^a A^{\mu b}, \quad (53)$$

where the mass matrix M_{ab} is given by

$$\frac{1}{2} M_{ab} = g^2 \text{Tr}[[T^a, v_3 T^3 + v_8 T^8][T^b, v_3 T^3 + v_8 T^8]]. \quad (54)$$

To find the masses of the gauge bosons, we need to diagonalize the mass matrix M_{ab} . Expressing the commutators in terms of the structure constants f^{abc} , we have

$$M_{ab} = 2g^2 \text{Tr}[[T^a, v_3 T^3 + v_8 T^8][T^b, v_3 T^3 + v_8 T^8]] \quad (55)$$

$$= 2g^2 \text{Tr}[(v_3 f^{a3c} T^c + v_8 f^{a8c} T^c)(v_3 f^{b3d} T^d + v_8 f^{b8d} T^d)] \quad (56)$$

$$= 2g^2 \text{Tr}[v_3^2 f^{a3c} f^{b3d} T^c T^d + v_8^2 f^{a8c} f^{b8d} T^c T^d + v_3 v_8 (f^{a3c} f^{b8d} + f^{a8c} f^{b3d}) T^c T^d] \quad (57)$$

$$= 2g^2 (v_3^2 f^{a3c} f^{b3d} + v_8^2 f^{a8c} f^{b8d} + v_3 v_8 (f^{a3c} f^{b8d} + f^{a8c} f^{b3d})) \text{Tr}(T^c T^d) \quad (58)$$

$$= g^2 (v_3^2 f^{a3c} f^{b3c} + v_8^2 f^{a8c} f^{b8c} + v_3 v_8 (f^{a3c} f^{b8c} + f^{a8c} f^{b3c})). \quad (59)$$

We just write down the structure constants f^{abc} of SU(3) here for reference:

$$f^{123} = 1, \quad f^{147} = -f^{156} = f^{246} = f^{257} = f^{345} = -f^{367} = \frac{1}{2}, \quad (60)$$

$$f^{458} = f^{678} = \frac{\sqrt{3}}{2}. \quad (61)$$

From the above structure constants, we can see that the mass matrix M_{ab} is diagonal, and the mass matrix becomes

$$(M_a)^2 = M_{aa} = g^2(v_3^2 f^{a3c} f^{b3c} + v_8^2 f^{a8c} f^{b8c} + v_3 v_8 (f^{a3c} f^{b8c} + f^{a8c} f^{b3c})) \quad (62)$$

$$= g^2 \left(v_3^2 f^{a3c} f^{a3c} + v_8^2 f^{a8c} f^{a8c} + v_3 v_8 (f^{a3c} f^{a8c} + f^{a8c} f^{a3c}) \right) \quad (63)$$

$$= g^2 (v_3^2 f^{a3c} f^{a3c} + v_8^2 f^{a8c} f^{a8c} + 2v_3 v_8 f^{a3c} f^{a8c}) \quad (64)$$

$$= g^2 (v_3 f^{a3c} + v_8 f^{a8c})^2 \quad (65)$$

Now we can calculate the masses of the gauge bosons by plugging in the values of f^{abc} for each a .

- For $a = 1$, we have $M_{11} = g^2(v_3 f^{13c} + v_8 f^{18c})^2 = g^2(v_3 f^{132} + v_8 f^{182})^2 = g^2 v_3^2$.
- For $a = 2$, we have $M_{22} = g^2(v_3 f^{23c} + v_8 f^{28c})^2 = g^2(v_3 f^{231} + v_8 f^{281})^2 = g^2 v_3^2$.
- For $a = 3$, we have $M_{33} = g^2(v_3 f^{33c} + v_8 f^{38c})^2 = 0$. Since $f^{33c} = 0$ and $f^{38c} = 0$ for all c , the gauge boson corresponding to $a = 3$ remains massless.
- For $a = 4$, we have $M_{44} = g^2(v_3 f^{43c} + v_8 f^{48c})^2 = g^2(v_3 f^{435} + v_8 f^{485})^2 = g^2 \left(-\frac{v_3}{2} - \frac{\sqrt{3}}{2} v_8 \right)^2$.
- For $a = 5$, we have $M_{55} = g^2(v_3 f^{53c} + v_8 f^{58c})^2 = g^2(v_3 f^{534} + v_8 f^{584})^2 = g^2 \left(\frac{v_3}{2} + \frac{\sqrt{3}}{2} v_8 \right)^2$.
- For $a = 6$, we have $M_{66} = g^2(v_3 f^{63c} + v_8 f^{68c})^2 = g^2(v_3 f^{637} + v_8 f^{687})^2 = g^2 \left(\frac{v_3}{2} - \frac{\sqrt{3}}{2} v_8 \right)^2$.
- For $a = 7$, we have $M_{77} = g^2(v_3 f^{73c} + v_8 f^{78c})^2 = g^2(v_3 f^{736} + v_8 f^{786})^2 = g^2 \left(-\frac{v_3}{2} + \frac{\sqrt{3}}{2} v_8 \right)^2$.
- For $a = 8$, we have $M_{88} = g^2(v_3 f^{83c} + v_8 f^{88c})^2 = 0$. Since $f^{83c} = 0$ and $f^{88c} = 0$ for all c , the gauge boson corresponding to $a = 8$ remains massless.

Thus, the masses of the gauge bosons are given by

$$M_1 = M_2 = g|v_3|, \quad (66)$$

$$M_3 = 0, \quad (67)$$

$$M_4 = M_5 = g \left| \frac{v_3}{2} + \frac{\sqrt{3}}{2} v_8 \right|, \quad (68)$$

$$M_6 = M_7 = g \left| \frac{v_3}{2} - \frac{\sqrt{3}}{2} v_8 \right|, \quad (69)$$

$$M_8 = 0. \quad (70)$$

In this general case, we have 6 massive gauge bosons and 2 massless gauge bosons. The masses of the gauge bosons depend on the values of v_3 and v_8 , which are constrained by the relation $v_3^2 + v_8^2 = v^2$. If we choose $v_3 = 0$ and $v_8 = v$, then the masses of the gauge bosons can be simplified to

$$M_1 = M_2 = 0, \quad (71)$$

$$M_3 = 0, \quad (72)$$

$$M_4 = M_5 = \frac{\sqrt{3}}{2} gv, \quad (73)$$

$$M_6 = M_7 = \frac{\sqrt{3}}{2} gv, \quad (74)$$

$$M_8 = 0. \quad (75)$$

Thus, in this case, we have 4 massive gauge bosons with mass $\frac{\sqrt{3}}{2} gv$ and 4 massless gauge bosons. If we choose $v_3 = v$ and $v_8 = 0$, then the masses of the gauge bosons can be simplified to

$$M_1 = M_2 = gv, \quad (76)$$

$$M_3 = 0, \quad (77)$$

$$M_4 = M_5 = \frac{gv}{2}, \quad (78)$$

$$M_6 = M_7 = \frac{gv}{2}, \quad (79)$$

$$M_8 = 0. \quad (80)$$

Thus, in this case, we have 2 massive gauge bosons with mass gv , 4 massive gauge bosons with mass $\frac{gv}{2}$, and 2 massless gauge bosons. Hence, we summarize the results as follows:

- (a) In the case $v_3 = 0$ and $v_8 = v$, the remaining unbroken gauge symmetry is $SU(2) \times U(1)$, where the $SU(2)$ subgroup is generated by T^1 , T^2 , and T^3 , and the $U(1)$ subgroup is generated by T^8 . The gauge bosons corresponding to the $SU(2)$ subgroup and the $U(1)$ subgroup remain massless. The massless bosons are A_μ^1 , A_μ^2 , A_μ^3 , and A_μ^8 . The massive bosons are A_μ^4 , A_μ^5 , A_μ^6 , and A_μ^7 with mass $\frac{\sqrt{3}}{2} gv$.
- (b) In the case $v_3 = v$ and $v_8 = 0$, the remaining unbroken gauge symmetry is $U(1) \times U(1)$, where one

U(1) subgroup is generated by T^3 and the other U(1) subgroup is generated by T^8 . The gauge bosons corresponding to both U(1) subgroups remain massless. The massless bosons are A_μ^3 and A_μ^8 . The massive bosons are A_μ^1 and A_μ^2 with mass gv , and A_μ^4 , A_μ^5 , A_μ^6 , and A_μ^7 with mass $\frac{gv}{2}$.

- (c) In the general case where both v_3 and v_8 are nonzero, if $v_3 \neq \pm\sqrt{3}v_8$, then the remaining unbroken gauge symmetry is $U(1) \times U(1)$. The gauge bosons corresponding to both U(1) subgroups remain massless. The massless bosons are A_μ^3 and A_μ^8 . The rest of the gauge bosons are massive with masses given by the above expressions. If $v_3 = \pm\sqrt{3}v_8$, then the remaining unbroken gauge symmetry is $SU(2) \times U(1)$. The gauge bosons corresponding to the SU(2) subgroup and the U(1) subgroup remain massless.

□

Homework 5 Due to February 26 8:00 AM

Question 1

Consider quantal one-dimensional problem described by the Lagrangian

$$\mathcal{L} = \frac{1}{2}\dot{X}^2 - V(X), \quad (81)$$

where

$$V(X) = \lambda(X^2 - \eta^2)^2, \quad (82)$$

where η is real and positive. This is so-called double-well potential. Find the classical trajectory in the Euclidean time connecting the point $X = -\eta$ in the distant past with the point $X = \eta$ in the distant future (keeping in mind that T will be set to ∞ at the end).

Why do we need Euclidean time? We will use this trajectory in subsequent lectures.

Answer

Now, the equation of motion in the Euclidean time is

$$\frac{d^2 X}{d\tau^2} = \frac{dV}{dX} = 4\lambda X(X^2 - \eta^2). \quad (83)$$

Note that $\tau = it$ is the Euclidean time. Now we can solve this equation by multiplying both sides by $dX/d\tau$:

$$\frac{dX}{d\tau} \frac{d^2 X}{d\tau^2} = 4\lambda X(X^2 - \eta^2) \frac{dX}{d\tau} \quad (84)$$

$$\implies \frac{1}{2} \frac{d}{d\tau} \left(\frac{dX}{d\tau} \right)^2 = 4\lambda X(X^2 - \eta^2) \frac{dX}{d\tau} = \frac{d}{d\tau} \lambda(X^2 - \eta^2)^2 \quad (85)$$

$$\implies \frac{1}{2} \left(\frac{dX}{d\tau} \right)^2 = \lambda(X^2 - \eta^2)^2 + C, \quad (86)$$

where C is an integration constant. Now, we can determine C by using the boundary condition. Since $X \rightarrow \pm\eta$ as $\tau \rightarrow \pm\infty$, we have

$$\lim_{\tau \rightarrow \pm\infty} \frac{1}{2} \left(\frac{dX}{d\tau} \right)^2 = \lim_{\tau \rightarrow \pm\infty} \lambda(X^2 - \eta^2)^2 + C = C. \quad (87)$$

On the other hand, since X is constant at $\tau \rightarrow \pm\infty$, we have $\lim_{\tau \rightarrow \pm\infty} dX/d\tau = 0$. Therefore, we have $C = 0$. Now, we can solve the equation

$$\frac{dX}{d\tau} = \sqrt{2\lambda}(X^2 - \eta^2). \quad (88)$$

This is a separable equation, and we can solve it by integrating both sides:

$$\int \frac{dX}{X^2 - \eta^2} = \int \sqrt{2\lambda} d\tau = \sqrt{2\lambda}\tau + C', \quad (89)$$

The left-hand side can be evaluated by partial fraction decomposition:

$$\int \frac{dX}{X^2 - \eta^2} = \int \frac{dX}{(X - \eta)(X + \eta)} = \frac{1}{2\eta} \int \left(\frac{1}{X - \eta} - \frac{1}{X + \eta} \right) dX = \frac{1}{2\eta} \ln \left| \frac{X - \eta}{X + \eta} \right|. \quad (90)$$

Therefore, we have

$$\frac{1}{2\eta} \ln \left| \frac{X - \eta}{X + \eta} \right| = \sqrt{2\lambda}\tau + C'. \quad (91)$$

Now, we can determine C' by using the boundary condition. Since $X \rightarrow -\eta$ as $\tau \rightarrow -\infty$, we have

$$\lim_{\tau \rightarrow -\infty} \frac{1}{2\eta} \ln \left| \frac{X - \eta}{X + \eta} \right| = \lim_{\tau \rightarrow -\infty} \sqrt{2\lambda}\tau + C' = -\infty + C' = -\infty. \quad (92)$$

We find this C' cannot be determined by the boundary condition. Without loss of generality, we can set $C' = \sqrt{2\lambda}\tau_0$ for some τ_0 . Then, we have

$$\frac{1}{2\eta} \ln \left| \frac{X - \eta}{X + \eta} \right| = \sqrt{2\lambda}(\tau - \tau_0). \quad (93)$$

Now, we can solve for X :

$$\ln \left| \frac{X - \eta}{X + \eta} \right| = 2\eta\sqrt{2\lambda}(\tau - \tau_0) \implies \left| \frac{X - \eta}{X + \eta} \right| = e^{2\eta\sqrt{2\lambda}(\tau - \tau_0)}. \quad (94)$$

Since $|X| < \eta$, we have $X + \eta > 0$ and $X - \eta < 0$, so we can drop the absolute value:

$$\frac{\eta - X}{X + \eta} = e^{2\eta\sqrt{2\lambda}(\tau - \tau_0)}. \quad (95)$$

Solving for X , we get

$$\eta - X = (X + \eta)e^{2\eta\sqrt{2\lambda}(\tau - \tau_0)}. \quad (96)$$

Rearranging terms, we have

$$\eta - X = X e^{2\eta\sqrt{2\lambda}(\tau-\tau_0)} + \eta e^{2\eta\sqrt{2\lambda}(\tau-\tau_0)}. \quad (97)$$

Collecting terms involving X , we get

$$X(1 + e^{2\eta\sqrt{2\lambda}(\tau-\tau_0)}) = \eta(1 - e^{2\eta\sqrt{2\lambda}(\tau-\tau_0)}). \quad (98)$$

Therefore,

$$X(\tau) = \frac{\eta(1 - e^{2\eta\sqrt{2\lambda}(\tau-\tau_0)})}{1 + e^{2\eta\sqrt{2\lambda}(\tau-\tau_0)}} = \eta \tanh(\eta\sqrt{2\lambda}(\tau - \tau_0)). \quad (99)$$

This is the classical trajectory in Euclidean time. The reason we need Euclidean time is that in Minkowski time, the trajectory connecting $X = -\eta$ and $X = \eta$ would be a solution to the equation of motion with negative kinetic energy, **which is not physical in classical mechanics sense**. In contrast, in Euclidean time, the kinetic energy is positive definite, allowing for a well-defined classical trajectory connecting the two minima of the potential. We can use this trajectory to compute the **tunneling phenomenon** between the two minima, which is a key aspect of quantum mechanics in this system.

Homework 6 Due to March 5 8:00 AM

Question 1

Consider the harmonic oscillator problem we have discussed previously at finite temperature T , i.e. on S^1 with circumference $\mathcal{T} = \beta = 1/T$ and Lagrangian

$$\mathcal{L} = \frac{1}{2}\dot{x}^2 - \frac{1}{2}\omega^2 x^2. \quad (100)$$

Impose the periodic boundary conditions and calculate the thermal partition function Z . What would you get with the anti-periodic boundary conditions?

Answer

First, we can expand the field $x(\tau)$ in Fourier modes as

$$x(\tau) = \sum_{n=-\infty}^{\infty} x_n e^{i\omega_n \tau}, \quad \omega_n = \frac{2\pi n}{\beta}. \quad (101)$$

In this way, we have imposed the periodic boundary conditions $x(\tau+\beta) = x(\tau)$, which leads to the quantization of the frequencies ω_n . Besides, the Fourier coefficients x_n are complex numbers satisfying $x_{-n} = x_n^*$ to ensure that $x(\tau)$ is real. Substituting this expansion into the action, the Euclidean action becomes (note that we have Wick rotated to Euclidean time $\tau = it$):

$$S_E = \int_0^\beta d\tau \left(\frac{1}{2}\dot{x}^2 + \frac{1}{2}\omega^2 x^2 \right) = \frac{1}{2} \int_0^\beta d\tau \sum_{n,m} \left(-\omega_n \omega_m x_n x_m e^{i(\omega_n + \omega_m)\tau} + \omega^2 x_n x_m e^{i(\omega_n + \omega_m)\tau} \right) \quad (102)$$

$$= \frac{1}{2} \sum_{n,m} (-\omega_n \omega_m + \omega^2) x_n x_m \int_0^\beta d\tau e^{i(\omega_n + \omega_m)\tau} = \frac{1}{2} \sum_{n,m} (-\omega_n \omega_m + \omega^2) x_n x_m \beta \delta_{n,-m} \quad (103)$$

$$= \frac{1}{2} \beta \sum_n (-\omega_n \omega_{-n} + \omega^2) x_n x_{-n} \quad (104)$$

$$= \frac{1}{2} \beta \sum_n (\omega_n^2 + \omega^2) |x_n|^2 = \frac{1}{2} \beta (\omega_0^2 + \omega^2) |x_0|^2 + \beta \sum_{n=1}^{\infty} (\omega_n^2 + \omega^2) |x_n|^2 \quad (105)$$

$$= \frac{1}{2} \beta (\omega_0^2 + \omega^2) x_0^2 + \beta \sum_{n=1}^{\infty} (\omega_n^2 + \omega^2) (a_n^2 + b_n^2), \quad (106)$$

where we have defined $x_n = a_n + ib_n$ with a_n, b_n being real numbers. The path integral measure can be written as

$$\mathcal{D}x = \prod_{n=-\infty}^{\infty} dx_n = dx_0 \prod_{n=1}^{\infty} da_n db_n. \quad (107)$$

Thus, the partition function can be computed as

$$Z = \int \mathcal{D}x e^{-S_E} = \int dx_0 e^{-\frac{1}{2}\beta(\omega_0^2 + \omega^2)x_0^2} \prod_{n=1}^{\infty} \int da_n db_n e^{-\beta(\omega_n^2 + \omega^2)(a_n^2 + b_n^2)} \quad (108)$$

$$= \int dx_0 e^{-\frac{1}{2}\beta(\omega_0^2 + \omega^2)x_0^2} \prod_{n=1}^{\infty} \left(\int da_n e^{-\beta(\omega_n^2 + \omega^2)a_n^2} \int db_n e^{-\beta(\omega_n^2 + \omega^2)b_n^2} \right). \quad (109)$$

We can evaluate the Gaussian integrals as

$$\int dx_0 e^{-\frac{1}{2}\beta(\omega_0^2 + \omega^2)x_0^2} = \sqrt{\frac{2\pi}{\beta(\omega_0^2 + \omega^2)}} = \sqrt{\frac{2\pi}{\beta\omega^2}}, \quad (110)$$

$$\int da_n e^{-\beta(\omega_n^2 + \omega^2)a_n^2} = \sqrt{\frac{\pi}{\beta(\omega_n^2 + \omega^2)}}, \quad \int db_n e^{-\beta(\omega_n^2 + \omega^2)b_n^2} = \sqrt{\frac{\pi}{\beta(\omega_n^2 + \omega^2)}}. \quad (111)$$

Therefore, the partition function can be expressed as

$$Z = \sqrt{\frac{2\pi}{\beta\omega^2}} \prod_{n=1}^{\infty} \left(\frac{\pi}{\beta(\omega_n^2 + \omega^2)} \right) = \sqrt{\frac{2\pi}{\beta\omega^2}} \prod_{n=1}^{\infty} \left(\frac{\pi}{\beta \left(\left(\frac{2\pi n}{\beta} \right)^2 + \omega^2 \right)} \right) \quad (112)$$

$$= \sqrt{\frac{2\pi}{\beta\omega^2}} \prod_{n=1}^{\infty} \left(\frac{\pi}{\beta \left(\frac{2\pi n}{\beta} \right)^2 \left(1 + \left(\frac{\beta\omega}{2\pi n} \right)^2 \right)} \right) \quad (113)$$

$$= \sqrt{\frac{2\pi}{\beta\omega^2}} \prod_{n=1}^{\infty} \left(\frac{\pi}{\beta \left(\frac{2\pi n}{\beta} \right)^2} \right) \prod_{n=1}^{\infty} \left(1 + \left(\frac{\beta\omega}{2\pi n} \right)^2 \right)^{-1} \quad (114)$$

To evaluate the infinite product, we can use the formula

$$\frac{\sinh(\pi z)}{\pi z} = \prod_{n=1}^{\infty} \left(1 + \frac{z^2}{n^2} \right) \quad (115)$$

$$\Rightarrow \frac{\pi z}{\sinh(\pi z)} = \prod_{n=1}^{\infty} \left(1 + \frac{z^2}{n^2} \right)^{-1}. \quad (116)$$

By setting $z = \frac{\beta\omega}{2\pi}$, we can rewrite the partition function as

$$Z = \sqrt{\frac{2\pi}{\beta\omega^2}} \left[\prod_{n=1}^{\infty} \left(\frac{\pi}{\beta \left(\frac{2\pi n}{\beta} \right)^2} \right) \right] \cdot \frac{\pi z}{\sinh(\pi z)} \quad (117)$$

$$= \sqrt{\frac{2\pi}{\beta\omega^2}} \left[\prod_{n=1}^{\infty} \left(\frac{\pi}{\beta \left(\frac{2\pi n}{\beta} \right)^2} \right) \right] \cdot \frac{\frac{\beta\omega}{2}}{\sinh\left(\frac{\beta\omega}{2}\right)} \quad (118)$$

$$= \sqrt{2\pi\beta} \left[\prod_{n=1}^{\infty} \left(\frac{\pi}{\beta \left(\frac{2\pi n}{\beta} \right)^2} \right) \right] \cdot \frac{1}{2 \sinh\left(\frac{\beta\omega}{2}\right)}. \quad (119)$$

We can further simplify the infinite product as

$$\prod_{n=1}^{\infty} \left(\frac{\pi}{\beta \left(\frac{2\pi n}{\beta} \right)^2} \right) = \prod_{n=1}^{\infty} \left(\frac{\beta}{4\pi n^2} \right) \quad (120)$$

Our final expression for the partition function is

$$Z(\beta, \omega) = \sqrt{2\pi\beta} \prod_{n=1}^{\infty} \left(\frac{\beta}{4\pi n^2} \right) \cdot \frac{1}{2 \sinh\left(\frac{\beta\omega}{2}\right)}. \quad (121)$$

Also, the last term can be rewritten as

$$\frac{1}{2 \sinh\left(\frac{\beta\omega}{2}\right)} = \frac{e^{-\frac{\beta\omega}{2}}}{1 - e^{-\beta\omega}} \quad (122)$$

$$= e^{-\frac{\beta\omega}{2}} \sum_{n=0}^{\infty} e^{-n\beta\omega} = \sum_{n=0}^{\infty} e^{-(n+\frac{1}{2})\beta\omega}. \quad (123)$$

This result is consistent with the energy spectrum of the quantum harmonic oscillator, which is given by $E_n = (n + \frac{1}{2})\omega$ with $n = 0, 1, 2, \dots$. The partition function can be expressed as

$$Z(\beta, \omega) = \sqrt{2\pi\beta} \prod_{n=1}^{\infty} \left(\frac{\beta}{4\pi n^2} \right) \sum_{n=0}^{\infty} e^{-(n+\frac{1}{2})\beta\omega}. \quad (124)$$

If we consider the normalized partition function by dividing Z by the infinite product, we have

$$\tilde{Z}(\beta, \omega) = \frac{Z(\beta, \omega)}{\sqrt{2\pi\beta} \prod_{n=1}^{\infty} \left(\frac{\beta}{4\pi n^2} \right)} = \sum_{n=0}^{\infty} e^{-(n+\frac{1}{2})\beta\omega}. \quad (125)$$

This is the standard expression for the partition function of a quantum harmonic oscillator at finite temperature.

For the anti-periodic boundary conditions, we can expand the field $x(\tau)$ in Fourier modes as

$$x(\tau) = \sum_{n=-\infty}^{\infty} x_{2n+1} e^{i\omega_{2n+1}\tau}, \quad \omega_{2n+1} = \frac{(2n+1)\pi}{\beta}. \quad (126)$$

In this case, the frequencies are quantized as odd multiples of π/β . Since $x(\tau)$ is anti-periodic, we have $x(\tau + \beta) = -x(\tau)$. Also, the Fourier coefficients x_{2n+1} are complex numbers satisfying $x_{-(2n+1)} = x_{2n+1}^*$ to ensure that $x(\tau)$ is real. Substituting this expansion into the Euclidean action, we have

$$S_E = \int_0^\beta d\tau \left(\frac{1}{2} \dot{x}^2 + \frac{1}{2} \omega^2 x^2 \right) \quad (127)$$

$$= \frac{1}{2} \int_0^\beta d\tau \sum_{n,m} \left(-\omega_{2n+1} \omega_{2m+1} x_{2n+1} x_{2m+1} e^{i(\omega_{2n+1} + \omega_{2m+1})\tau} + \omega^2 x_{2n+1} x_{2m+1} e^{i(\omega_{2n+1} + \omega_{2m+1})\tau} \right) \quad (128)$$

$$= \frac{1}{2} \sum_{n,m} \left(-\omega_{2n+1} \omega_{2m+1} + \omega^2 \right) x_{2n+1} x_{2m+1} \int_0^\beta d\tau e^{i(\omega_{2n+1} + \omega_{2m+1})\tau} \quad (129)$$

$$= \frac{1}{2} \sum_{n,m} \left(-\omega_{2n+1} \omega_{2m+1} + \omega^2 \right) x_{2n+1} x_{2m+1} \beta \delta_{n,-m-1} \quad (130)$$

$$= \frac{1}{2} \beta \sum_n \left(-\omega_{2n+1} \omega_{-2n-1} + \omega^2 \right) x_{2n+1} x_{-2n-1} = \frac{1}{2} \beta \sum_n \left(\omega_{2n+1}^2 + \omega^2 \right) |x_{2n+1}|^2 \quad (131)$$

$$= \beta \sum_{n=-\infty}^{\infty} \left(\omega_{2n+1}^2 + \omega^2 \right) (a_n^2 + b_n^2), \quad (132)$$

where we have defined $x_{2n+1} = a_n + ib_n$ with a_n, b_n being real numbers. The path integral measure can be written as

$$\mathcal{D}x = \prod_{n=-\infty}^{\infty} dx_{2n+1} = \prod_{n=-\infty}^{\infty} da_n db_n. \quad (133)$$

Thus, the partition function can be computed as

$$Z = \int \mathcal{D}x e^{-S_E} = \prod_{n=-\infty}^{\infty} \int da_n db_n e^{-\beta(\omega_{2n+1}^2 + \omega^2)(a_n^2 + b_n^2)} = \prod_{n=-\infty}^{\infty} \left(\int da_n e^{-\beta(\omega_{2n+1}^2 + \omega^2)a_n^2} \int db_n e^{-\beta(\omega_{2n+1}^2 + \omega^2)b_n^2} \right). \quad (134)$$

We can evaluate the Gaussian integrals as

$$\int da_n e^{-\beta(\omega_{2n+1}^2 + \omega^2)a_n^2} = \sqrt{\frac{\pi}{\beta(\omega_{2n+1}^2 + \omega^2)}}, \quad \int db_n e^{-\beta(\omega_{2n+1}^2 + \omega^2)b_n^2} = \sqrt{\frac{\pi}{\beta(\omega_{2n+1}^2 + \omega^2)}}. \quad (135)$$

Therefore, the partition function can be expressed as

$$Z = \prod_{n=-\infty}^{\infty} \left(\frac{\pi}{\beta(\omega_{2n+1}^2 + \omega^2)} \right) = \prod_{n=-\infty}^{\infty} \left(\frac{\pi}{\beta \left(\left(\frac{(2n+1)\pi}{\beta} \right)^2 + \omega^2 \right)} \right) \quad (136)$$

$$= \prod_{n=-\infty}^{\infty} \left(\frac{\pi}{\beta \left(\frac{(2n+1)\pi}{\beta} \right)^2 \left(1 + \left(\frac{\beta\omega}{(2n+1)\pi} \right)^2 \right)} \right) \quad (137)$$

$$= \prod_{n=-\infty}^{\infty} \left(\frac{\pi}{\beta \left(\frac{(2n+1)\pi}{\beta} \right)^2} \right) \prod_{n=-\infty}^{\infty} \left(1 + \left(\frac{\beta\omega}{(n + \frac{1}{2})2\pi} \right)^2 \right)^{-1} \quad (138)$$

To evaluate the infinite product, we can use the formula

$$\frac{\cosh(\pi z)}{\pi z} = \prod_{n=0}^{\infty} \left(1 + \frac{z^2}{(n + \frac{1}{2})^2} \right) \quad (139)$$

$$\Rightarrow \frac{\pi z}{\cosh(\pi z)} = \prod_{n=0}^{\infty} \left(1 + \frac{z^2}{(n + \frac{1}{2})^2} \right)^{-1}. \quad (140)$$

By setting $z = \frac{\beta\omega}{2\pi}$, we can rewrite the partition function as

$$Z = \prod_{n=-\infty}^{\infty} \left(\frac{\pi}{\beta \left(\frac{(2n+1)\pi}{\beta} \right)^2} \right) \cdot \frac{\pi z}{\cosh(\pi z)} \quad (141)$$

$$= \prod_{n=-\infty}^{\infty} \left(\frac{\pi}{\beta \left(\frac{(2n+1)\pi}{\beta} \right)^2} \right) \cdot \frac{\frac{\beta\omega}{2}}{\cosh\left(\frac{\beta\omega}{2}\right)} \quad (142)$$

$$= \prod_{n=-\infty}^{\infty} \left(\frac{\pi}{\beta \left(\frac{(2n+1)\pi}{\beta} \right)^2} \right) \cdot \frac{1}{2 \cosh\left(\frac{\beta\omega}{2}\right)}. \quad (143)$$

We can further simplify the infinite product as

$$\prod_{n=-\infty}^{\infty} \left(\frac{\pi}{\beta \left(\frac{(2n+1)\pi}{\beta} \right)^2} \right) = \prod_{n=-\infty}^{\infty} \left(\frac{\beta}{4\pi(n + \frac{1}{2})^2} \right). \quad (144)$$

Thus, our final expression for the partition function is

$$Z(\beta, \omega) = \prod_{n=-\infty}^{\infty} \left(\frac{\beta}{4\pi(n + \frac{1}{2})^2} \right) \cdot \frac{1}{2 \cosh\left(\frac{\beta\omega}{2}\right)}. \quad (145)$$

Also, the last term can be rewritten as

$$\frac{1}{2 \cosh\left(\frac{\beta\omega}{2}\right)} = \frac{e^{-\frac{\beta\omega}{2}}}{1 + e^{-\beta\omega}} \quad (146)$$

$$= e^{-\frac{\beta\omega}{2}} \sum_{n=0}^{\infty} (-1)^n e^{-n\beta\omega} = \sum_{n=0}^{\infty} (-1)^n e^{-(n+\frac{1}{2})\beta\omega}. \quad (147)$$

This result is consistent with the energy spectrum of the quantum harmonic oscillator with anti-periodic boundary conditions, which is given by $E_n = (n + \frac{1}{2})\omega$ with $n = 0, 1, 2, \dots$. The partition function can be expressed as

$$Z(\beta, \omega) = \prod_{n=-\infty}^{\infty} \left(\frac{\beta}{4\pi(n + \frac{1}{2})^2} \right) \sum_{n=0}^{\infty} (-1)^n e^{-(n+\frac{1}{2})\beta\omega}. \quad (148)$$

If we consider the normalized partition function by dividing Z by the infinite product, we have

$$\tilde{Z}(\beta, \omega) = \frac{Z(\beta, \omega)}{\prod_{n=-\infty}^{\infty} \left(\frac{\beta}{4\pi(n + \frac{1}{2})^2} \right)} = \sum_{n=0}^{\infty} (-1)^n e^{-(n+\frac{1}{2})\beta\omega}. \quad (149)$$

This is the standard expression for the partition function of a quantum harmonic oscillator with anti-periodic boundary conditions at finite temperature. $\square$

Question 2

Using the solution of the previous problem determine the free energy F , entropy S and average energy E in terms of the thermal partition function.

Answer

The free energy F can be calculated from the partition function Z as

$$F = -\frac{1}{\beta} \ln Z. \quad (150)$$

The average energy E can be calculated from the partition function as

$$E = -\frac{\partial}{\partial \beta} \ln Z. \quad (151)$$

The entropy S can be calculated from the free energy and average energy as

$$S = \beta(E - F). \quad (152)$$

By substituting the expression for F into the formula for S , we can express the entropy in terms of the partition function as

$$S = \beta \left(E + \frac{1}{\beta} \ln Z \right) = \beta E + \ln Z. \quad (153)$$

Thus, we have expressed the free energy, average energy, and entropy in terms of the thermal partition function Z .

Now, we can further express these thermodynamic quantities in terms of the normalized partition function $\tilde{Z}$ and we drop the tilde for simplicity. For the periodic boundary conditions, we have

$$Z(\beta, \omega) = \frac{1}{2 \sinh \left(\frac{\beta \omega}{2} \right)} = \frac{e^{-\frac{\beta \omega}{2}}}{1 - e^{-\beta \omega}}, \quad (154)$$

and for the anti-periodic boundary conditions, we have

$$Z(\beta, \omega) = \frac{1}{2 \cosh \left(\frac{\beta \omega}{2} \right)} = \frac{e^{-\frac{\beta \omega}{2}}}{1 + e^{-\beta \omega}}. \quad (155)$$

By substituting these expressions for Z into the formulas for F , E , and S , we can obtain the explicit expressions for these thermodynamic quantities in terms of β and ω for both cases. For the periodic boundary conditions,

we have

$$F = -\frac{1}{\beta} \ln \left(\frac{e^{-\frac{\beta\omega}{2}}}{1 - e^{-\beta\omega}} \right) = \frac{\omega}{2} + \frac{1}{\beta} \ln (1 - e^{-\beta\omega}), \quad (156)$$

$$E = -\frac{\partial}{\partial\beta} \ln \left(\frac{e^{-\frac{\beta\omega}{2}}}{1 - e^{-\beta\omega}} \right) = \frac{\omega}{2} + \frac{\omega e^{-\beta\omega}}{1 - e^{-\beta\omega}}, \quad (157)$$

$$S = \beta E + \ln \left(\frac{e^{-\frac{\beta\omega}{2}}}{1 - e^{-\beta\omega}} \right) = \frac{\beta\omega}{2} + \frac{\beta\omega e^{-\beta\omega}}{1 - e^{-\beta\omega}} + \ln \left(\frac{e^{-\frac{\beta\omega}{2}}}{1 - e^{-\beta\omega}} \right) \quad (158)$$

$$= \frac{\beta\omega}{2} + \frac{\beta\omega e^{-\beta\omega}}{1 - e^{-\beta\omega}} - \frac{\beta\omega}{2} - \ln (1 - e^{-\beta\omega}) \quad (159)$$

$$= \frac{\beta\omega e^{-\beta\omega}}{1 - e^{-\beta\omega}} - \ln (1 - e^{-\beta\omega}). \quad (160)$$

For the anti-periodic boundary conditions, we have

$$F = -\frac{1}{\beta} \ln \left(\frac{e^{-\frac{\beta\omega}{2}}}{1 + e^{-\beta\omega}} \right) = \frac{\omega}{2} + \frac{1}{\beta} \ln (1 + e^{-\beta\omega}), \quad (161)$$

$$E = -\frac{\partial}{\partial\beta} \ln \left(\frac{e^{-\frac{\beta\omega}{2}}}{1 + e^{-\beta\omega}} \right) = \frac{\omega}{2} - \frac{\omega e^{-\beta\omega}}{1 + e^{-\beta\omega}}, \quad (162)$$

$$S = \beta E + \ln \left(\frac{e^{-\frac{\beta\omega}{2}}}{1 + e^{-\beta\omega}} \right) = \frac{\beta\omega}{2} - \frac{\beta\omega e^{-\beta\omega}}{1 + e^{-\beta\omega}} + \ln \left(\frac{e^{-\frac{\beta\omega}{2}}}{1 + e^{-\beta\omega}} \right) \quad (163)$$

$$= \frac{\beta\omega}{2} - \frac{\beta\omega e^{-\beta\omega}}{1 + e^{-\beta\omega}} - \frac{\beta\omega}{2} - \ln (1 + e^{-\beta\omega}) \quad (164)$$

$$= -\frac{\beta\omega e^{-\beta\omega}}{1 + e^{-\beta\omega}} - \ln (1 + e^{-\beta\omega}). \quad (165)$$

□

Homework 7 Due to March 19 8:00 AM

Question 1

Compare the composition of the Dirac spinor and all five gamma matrices, including

$$\gamma^5 = \pm i \prod_{\mu=0,1,2,3} \gamma^\mu, \quad (166)$$

in Peskin and Schroeder, Landau-Lifshitz's Quantum Electrodynamics, and with the one introduced in my course. Find the unitary transition matrix U which implements the transition from the spinorial to the standard (Dirac) representation.

Answer

In Peskin and Schroeder, the gamma matrices are defined as

$$\gamma^0 = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}, \quad \gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} -I & 0 \\ 0 & I \end{pmatrix}, \quad (167)$$

where σ^i are the Pauli matrices. In Landau-Lifshitz's Quantum Electrodynamics, the gamma matrices are defined as

$$\gamma^0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}, \quad \gamma^5 = -i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} 0 & -I \\ -I & 0 \end{pmatrix}. \quad (168)$$

In the course, the gamma matrices are defined as

$$\gamma^0 = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}, \quad \gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} -I & 0 \\ 0 & I \end{pmatrix}. \quad (169)$$

The unitary transition matrix U that implements the transition from the spinorial to the standard (Dirac) representation can be found by comparing the definitions of the gamma matrices in the two representations. We can write the gamma matrices in the spinorial representation as

$$\gamma_{\text{spinorial}}^0 = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}, \quad \gamma_{\text{spinorial}}^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}, \quad \gamma_{\text{spinorial}}^5 = i\gamma_{\text{spinorial}}^0\gamma_{\text{spinorial}}^1\gamma_{\text{spinorial}}^2\gamma_{\text{spinorial}}^3 = \begin{pmatrix} -I & 0 \\ 0 & I \end{pmatrix}. \quad (170)$$

The gamma matrices in the standard (Dirac) representation are

$$\gamma_{\text{Dirac}}^0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \quad \gamma_{\text{Dirac}}^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}, \quad \gamma_{\text{Dirac}}^5 = i\gamma_{\text{Dirac}}^0\gamma_{\text{Dirac}}^1\gamma_{\text{Dirac}}^2\gamma_{\text{Dirac}}^3 = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}. \quad (171)$$

To find the unitary transition matrix U , we can use the fact that the gamma matrices in the two representations are related by a similarity transformation:

$$\gamma_{\text{Dirac}}^\mu = U \gamma_{\text{spinorial}}^\mu U^\dagger. \quad (172)$$

By comparing the definitions of the gamma matrices in the two representations, we can find that the unitary transition matrix U is given by

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} I & I \\ -I & I \end{pmatrix}. \quad (173)$$

This matrix satisfies the relation

$$U \gamma_{\text{spinorial}}^0 U^\dagger = \gamma_{\text{Dirac}}^0, \quad U \gamma_{\text{spinorial}}^i U^\dagger = \gamma_{\text{Dirac}}^i, \quad U \gamma_{\text{spinorial}}^5 U^\dagger = \gamma_{\text{Dirac}}^5. \quad (174)$$

□

Question 2

Compare the definitions of left-handed and right-handed. As a reference point use the Dirac equation of motion

$$i\gamma^\mu\partial_\mu\psi = m\psi \quad (175)$$

and the fact that neutrino is left-handed. Check that in all definitions of the Dirac matrices the Clifford algebra is satisfied.

Answer

We can define the Dirac spinor ψ as a four-component with a left-handed Weyl spinor ψ_L and a right-handed Weyl spinor ψ_R , which have 2 components each:

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}. \quad (176)$$

The left-handed and right-handed components of the Dirac spinor can be defined using the projection operators:

$$P_L = \frac{1}{2}(1 - \gamma^5), \quad P_R = \frac{1}{2}(1 + \gamma^5). \quad (177)$$

The left-handed and right-handed components of the Dirac spinor can be obtained by applying the projection operators to the Dirac spinor:

$$\psi_L = P_L\psi, \quad \psi_R = P_R\psi. \quad (178)$$

We can write the Dirac equation of motion in terms of the left-handed and right-handed components:

$$i \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix} \begin{pmatrix} \partial_\mu\psi_L \\ \partial_\mu\psi_R \end{pmatrix} = m \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}, \quad (179)$$

where $\sigma^\mu = (I, \sigma^i)$ and $\bar{\sigma}^\mu = (I, -\sigma^i)$. This equation can be separated into two equations:

$$i\sigma^\mu\partial_\mu\psi_R = m\psi_L, \quad i\bar{\sigma}^\mu\partial_\mu\psi_L = m\psi_R. \quad (180)$$

In the massless limit ($m = 0$), the equations decouple and we have

$$i\sigma^\mu\partial_\mu\psi_R = 0, \quad i\bar{\sigma}^\mu\partial_\mu\psi_L = 0. \quad (181)$$

This shows that the left-handed and right-handed components of the Dirac spinor are independent in the massless limit. Hence, the massless fermion can be left-handed or right-handed. Hence, the neutrino, which is observed to be left-handed, can be described by a left-handed Weyl spinor ψ_L in the massless limit.

To check that the Clifford algebra is satisfied in all definitions of the Dirac matrices, we need to verify that the gamma matrices satisfy the anticommutation relation:

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}I, \quad (182)$$

where $g^{\mu\nu}$ is the Minkowski metric. Here we use the Weyl representation of the gamma matrices, which is given by

$$\gamma^0 = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}, \quad \gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} -I & 0 \\ 0 & I \end{pmatrix}. \quad (183)$$

Then we can compute the anticommutators:

$$\{\gamma^0, \gamma^0\} = 2I, \quad (184)$$

$$\{\gamma^i, \gamma^j\} = -2\delta^{ij}I, \quad (185)$$

$$\{\gamma^0, \gamma^i\} = 0, \quad (186)$$

$$\{\gamma^5, \gamma^\mu\} = 0. \quad (187)$$

Here we apply the properties of the Pauli matrices, which satisfy $\{\sigma^i, \sigma^j\} = 2\delta^{ij}I$. Since we can apply the unitary transition matrix U to transform the gamma matrices to the standard (Dirac) representation, we can conclude that the Clifford algebra is satisfied in all definitions of the Dirac matrices. $\square$

Homework 8 Due to March 26 8:00 AM

Question a

Assume you have a static point-like electric charge e at the point y . Find the time component of the four-potential $A_0(t, \vec{x})$ in the Lorentz-Heaviside system of units.

Answer

The potential $A_0(t, \vec{x})$ due to a static point-like electric charge e at the point $\vec{y}$ can be found using the Green's function for the Poisson equation in three-dimensional space. The Poisson equation for the electrostatic potential is given by

$$\nabla^2 A_0(\vec{x}) = -\rho(\vec{x}), \quad (188)$$

where $\rho(\vec{x})$ is the charge density. For a point charge e located at $\vec{y}$, the charge density can be expressed as

$$\rho(\vec{x}) = e\delta^3(\vec{x} - \vec{y}), \quad (189)$$

where δ^3 is the three-dimensional Dirac delta function. The Green's function for the Poisson equation in three dimensions is given by

$$G(\vec{x}, \vec{y}) = \frac{1}{4\pi|\vec{x} - \vec{y}|}. \quad (190)$$

Using the Green's function, we can express the potential $A_0(\vec{x})$ as

$$A_0(\vec{x}) = \int d^3y' G(\vec{x}, \vec{y}') \rho(\vec{y}'). \quad (191)$$

Substituting the charge density for the point charge, we have

$$A_0(\vec{x}) = \int d^3y' \frac{1}{4\pi|\vec{x} - \vec{y}'|} e\delta^3(\vec{y}' - \vec{y}) = \frac{e}{4\pi|\vec{x} - \vec{y}|}. \quad (192)$$

Thus, the time component of the four-potential $A_0(t, \vec{x})$ in the Lorentz-Heaviside system of units is given by

$$A_0(t, \vec{x}) = \frac{e}{4\pi|\vec{x} - \vec{y}|}. \quad (193)$$

□

Question b

Use the previous calculation as a hint to obtain the photon propagator (Green's function) in the Coulomb gauge $\partial_i A_i = 0$ (or in the momentum space $k_i A_i = 0$). Start from the Feynman gauge

$$G^{\mu\nu}(k) = \frac{-ig^{\mu\nu}}{k^2}. \quad (194)$$

Answer

In the Coulomb gauge, we impose the condition $\partial_i A_i = 0$ in position space, which translates to $k_i A_i = 0$ in momentum space. This means that the spatial components of the photon propagator must be transverse to the momentum vector $\vec{k}$. To achieve this, we can decompose the Feynman gauge propagator into its time and spatial components. The Feynman gauge propagator is given by

$$G^{\mu\nu}(k) = \frac{-ig^{\mu\nu}}{k^2}. \quad (195)$$

The projection operator that enforces the Coulomb gauge condition can be defined as

$$P^{ij} = \delta^{ij} - \frac{k^i k^j}{\vec{k}^2}, \quad (196)$$

where i, j run over the spatial indices. The time component of the propagator remains unchanged, while the spatial components are modified by the projection operator. Thus, the photon propagator in the Coulomb gauge can be expressed as

$$G^{00}(k) = \frac{i}{\vec{k}^2}, \quad (197)$$

$$G^{0i}(k) = G^{i0}(k) = 0, \quad (198)$$

$$G^{ij}(k) = \frac{i}{k^2} \left(\delta^{ij} - \frac{k^i k^j}{\vec{k}^2} \right) = \frac{iP^{ij}}{k^2}. \quad (199)$$

□

Homework 9 Due to April 2 8:00 AM

Question 1

Repeat the Faddeev-Popov-'t Hooft quantization procedure that we have studied previously for pure Yang-Mills theory. Find an appropriate gauge condition such that it cancels the quadratic cross term in Equation:

$$-ieA^\mu(\phi^\dagger \overleftrightarrow{\partial}_\mu \phi) = evA_\mu \partial^\mu \varphi + \text{cubic terms.} \quad (200)$$

Collect all quadratic terms in the gauge-fixed Lagrangian (with ghosts included) and determine the propagators of the fields A_μ, h, c and ϕ as function of ξ . In thus obtained propagators study separately the limits

$$\xi = 1 \text{ and } \xi \rightarrow \infty. \quad (201)$$

What information can you obtain from the limit $\xi \rightarrow \infty$?

Answer

First, we have to identify the gauge-fixing term. For a U(1) gauge theory coupled to a scalar field, we choose the gauge condition:

$$\partial_\mu A^\mu - \xi ev\varphi = 0 \quad (202)$$

This Lorenz-type gauge with the scalar field term cancels the cross term. The gauge-fixing Lagrangian is:

$$\mathcal{L}_{\text{gf}} = -\frac{1}{2\xi}(\partial_\mu A^\mu - \xi ev\varphi)^2 \quad (203)$$

$$= -\frac{1}{2\xi}(\partial^\mu A_\mu)^2 + ev\varphi(\partial^\mu A_\mu) - \frac{\xi}{2}e^2v^2\varphi^2 \quad (204)$$

$$= -\frac{1}{2\xi}(\partial^\mu A_\mu)^2 - evA_\mu(\partial^\mu \varphi) - \frac{\xi}{2}e^2v^2\varphi^2, \quad (205)$$

where we apply the IBP to move the derivative to act on φ . Define $m_A = ev$, then obtain:

$$\mathcal{L}_{\text{gf}} = -\frac{1}{2\xi}(\partial^\mu A_\mu)^2 - m_A A_\mu(\partial^\mu \varphi) - \frac{\xi}{2}m_A^2\varphi^2. \quad (206)$$

The ghost lagrangian is derived from the gauge fixing term, which is given by:

$$\delta_\alpha G = \partial^\mu(\delta A_\mu) - \xi ev(\delta\varphi) \quad (207)$$

$$= \partial^\mu \left(\frac{1}{e} \partial_\mu \alpha \right) - \xi ev(\alpha(v+h)) \quad (208)$$

$$= \frac{1}{e} [\partial^2 - \xi e^2 v(v+h)] \alpha. \quad (209)$$

If we only consider the quadratic term and ignore the h , then the lagrangian is given by:

$$\mathcal{L}_{\text{ghost}} = \bar{c} [-\partial^2 - \xi e^2 v^2] c \quad (210)$$

$$= \bar{c} [-\partial^2 - \xi m_A^2] c. \quad (211)$$

Also, after SSB, higgs field with VEV become

$$|D_\mu \phi|^2 \supset \frac{1}{2}(\partial_\mu \varphi)^2 + ev A_\mu \partial^\mu \varphi + \frac{1}{2}(ev)^2 A_\mu A^\mu = \frac{1}{2}(\partial_\mu \varphi)^2 + m_A A_\mu \partial^\mu \varphi + \frac{1}{2}m_A^2 A_\mu A^\mu \quad (212)$$

Now, the full lagrangian with the quadratic terms is given by

$$\mathcal{L}_{\text{full}} = \mathcal{L}_{\text{higgs with VEV}} + \mathcal{L}_{\text{U(1)}} + \mathcal{L}_{\text{gf}} + \mathcal{L}_{\text{ghost}} \quad (213)$$

$$= \frac{1}{2}(\partial_\mu h)^2 - \frac{1}{2}m_h^2 h^2 + \frac{1}{2}(\partial_\mu \varphi)^2 + \cancel{m_A A_\mu \partial^\mu \varphi} + \frac{1}{2}m_A^2 A_\mu A^\mu - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial^\mu A_\mu)^2 - \cancel{m_A A_\mu (\partial^\mu \varphi)} \quad (214)$$

$$- \frac{\xi}{2}m_A^2 \varphi^2 + \bar{c} [-\partial^2 - \xi m_A^2] c \quad (215)$$

$$= \frac{1}{2}(\partial_\mu h)^2 - \frac{1}{2}m_h^2 h^2 + \frac{1}{2}(\partial_\mu \varphi)^2 + \frac{1}{2}m_A^2 A_\mu A^\mu - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial^\mu A_\mu)^2 - \frac{\xi}{2}m_A^2 \varphi^2 + \bar{c} [-\partial^2 - \xi m_A^2] c \quad (216)$$

$$= \underbrace{\frac{1}{2}h(-\partial^2 - m_h^2)h}_{\text{physical higgs}} \quad (217)$$

$$- \underbrace{\frac{1}{2}A_\mu \left[(-\partial^2 + m_A^2)g^{\mu\nu} + \left(1 - \frac{1}{\xi}\right) \partial^\mu \partial^\nu \right] A_\nu}_{\text{Gauge}} \quad (218)$$

$$+ \underbrace{\frac{1}{2}\varphi(-\partial^2 - \xi m_A^2)\varphi}_{\text{Goldstone boson}} \quad (219)$$

$$+ \underbrace{\bar{c}(-\partial^2 - \xi m_A^2)c}_{\text{ghost}} \quad (220)$$

The corresponding propagators are given by:

- physical higgs:

$$\Delta(k) = \frac{i}{k^2 - m_h^2}. \quad (221)$$

- Photon:

$$\Delta_{\mu\nu}(k) = \frac{i}{k^2 - m_A^2} \left(-g_{\mu\nu} + \frac{k_\mu k_\nu}{k^2 - \xi m_A^2} (1 - \xi) \right). \quad (222)$$

- Goldstone boson:

$$\Delta(k) = \frac{i}{k^2 - \xi m_A^2}. \quad (223)$$

- Ghost:

$$\Delta(k) = \frac{i}{k^2 - \xi m_A^2}. \quad (224)$$

Now, if we consider the $\xi = 1$, we have

- physical higgs:

$$\Delta(k) = \frac{i}{k^2 - m_h^2}. \quad (225)$$

- Photon:

$$\Delta_{\mu\nu}(k) = \frac{-ig_{\mu\nu}}{k^2 - m_A^2}. \quad (226)$$

- Goldstone boson:

$$\Delta(k) = \frac{i}{k^2 - m_A^2}. \quad (227)$$

- Ghost:

$$\Delta(k) = \frac{i}{k^2 - m_A^2}. \quad (228)$$

If we consider $\xi \rightarrow \infty$, we have

- physical higgs:

$$\Delta(k) = \frac{i}{k^2 - m_h^2}. \quad (229)$$

- Photon:

$$\Delta_{\mu\nu}(k) = \frac{i}{k^2 - m_A^2} \left(-g_{\mu\nu} + \frac{k_\mu k_\nu}{k^2 - \xi m_A^2} (1 - \xi) \right) \rightarrow \frac{-i}{k^2 - m_A^2} \left(g_{\mu\nu} - \frac{k_\mu k_\nu}{m_A^2} \right). \quad (230)$$

- Goldstone boson:

$$\Delta(k) = \frac{i}{k^2 - \xi m_A^2} \rightarrow 0. \quad (231)$$

- Ghost:

$$\Delta(k) = \frac{i}{k^2 - \xi m_A^2} \rightarrow 0. \quad (232)$$

Interpretation: When we take the limit $\xi \rightarrow \infty$, the photon can eat the goldstone boson and ghost to physically become massive vector boson. Also, under this limit, there is no non-physical field c and non-physical parameter ξ in the calculation.

Homework 10 Due to April 11 8:00 AM

Question 1

Assume we have Yang-Mills theory with the gauge sector

$$G = SU(2), \quad (233)$$

and the Higgs sector consisting of one real scalar field in the *adjoint* representation of $SU(2)$.

The adjoint field develops a vacuum expectation value $\langle \phi^a \phi^a \rangle = v^2$. Calculate the first coefficient in the gauge coupling β function for $\mu \gg M_W, M_H$. Compare the result to the first coefficient in the β function of pure Yang-Mills theory with the same gauge group as above. Can you give a physical interpretation for the difference?

Assume that $M_W \ll M_H$. Find the first coefficients in the β function in the window $M_W \ll \mu \ll M_H$.

Answer

We quote the equation (16.133) in Peskin:

$$\beta(g) = -\frac{g^3}{(4\pi)^2} \left(\frac{11}{3} C_2(G) - \frac{4}{3} n_f C(r) \right). \quad (234)$$

However, we can generalize this equation by including the Higgs sector. For a theory with scalars in the representation r_s , the first coefficient b_0 of the β function (where $\beta(g) = -\frac{b_0}{16\pi^2} g^3$) is given by:

$$b_0 = \frac{11}{3} C_2(G) - \sum_s \frac{1}{3} C(r_s) \cdot (\text{const}) - \sum_f \frac{4}{3} S_2(r_f) \quad (235)$$

$$= \frac{11}{3} C_2(G) - \sum_s \frac{1}{3} C(r_s) \cdot (\text{const}) \quad (236)$$

where $C(r_s)$ is the index of the representation r_s (often denoted as $T(r_s)$), while the spin-dependent factors are already included in the prefactors of the formula. For a real scalar field in the adjoint representation of $G = SU(2)$, we have:

- $C_2(G) = 2$.
- $C(\text{adj}) = C_2(G) = 2$.
- For a real scalar, the contribution is half of that of a complex scalar. Thus, the coefficient is $1/6$ instead of $1/3$ in the standard complex scalar formula. More specifically, the const in the equation now is $\frac{1}{2}$.

(a) $\mu \gg M_W, M_H$:

In this regime, we can treat all fields in the theory as effectively massless. Both the gauge bosons and the adjoint scalar field contribute to the vacuum polarization. Using the values identified above, the first coefficient b_0 is:

$$b_0^{\text{high}} = \frac{11}{3}(2) - \frac{1}{6}(2) = \frac{22}{3} - \frac{1}{3} = \frac{21}{3} = 7. \quad (237)$$

(b) $M_W \ll \mu \ll M_H$:

Since we assume $M_W \ll \mu \ll M_H$, the scalar field (with mass M_H) has "frozen out" or decoupled from the renormalization group flow. However, the gauge bosons remain active since μ is still much larger than their mass scale M_W . Therefore, the theory effectively behaves like a pure $SU(2)$ Yang-Mills theory in this regime. The first coefficient is:

$$b_0^{\text{mid}} = \frac{11}{3}C_2(G) = \frac{11}{3}(2) = \frac{22}{3}. \quad (238)$$

In this window, the coupling $g(\mu)$ runs faster (the beta function is more negative) than in the high-energy regime because the screening contribution from the adjoint scalar is no longer present.

Physical Interpretation of the difference: Comparing $b_0^{\text{high}} = 7$ to $b_0^{\text{mid}} = 22/3$, we see that the presence of the adjoint scalar field decreases the value of b_0 . In Yang-Mills theory, the gauge boson self-interactions lead to anti-screening (asymptotic freedom), which corresponds to the $11/3$ term. On the other hand, the scalar field contributes to screening (the QED-like behavior). Therefore, adding an adjoint scalar field increases the screening effect in the UV, which makes the theory "less" asymptotically free compared to the pure gauge theory.

Homework 11 Due to April 16 8:00 AM

Question 1

In two-dimensional Schwinger model (massless 2D QED)

$$\mathcal{L} = -\frac{1}{4e_0^2} F_{\mu\nu} F^{\mu\nu} + \bar{\psi} i \not{D} \psi, \quad (239)$$

where

$$D_\mu = \partial_\mu + i A_\mu, \quad (240)$$

and ψ is a two-dimensional Dirac spinor. Consider the axial-vector current,

$$j_\mu^5 = \bar{\psi} \gamma_\mu \gamma_5 \psi. \quad (241)$$

Calculate $\partial^\mu j_\mu^5$ by virtue of the classical equations of motion for ψ , $\bar{\psi}$.

Regularize (the UV divergent) current j_μ^5 by adding the Pauli-Villars regulator field Ψ_R

$$\mathcal{L}_R = \bar{\Psi}_R (i \not{D} - M) \Psi_R, \quad \mathcal{L}_{\text{total}} = \mathcal{L} + \mathcal{L}_R, \quad (242)$$

which is similar to ψ in everything except two aspects: (i) it has a heavy mass M ($M \rightarrow \infty$ at the end of the calculation); (ii) every loop of the primary ψ field is accompanied by the similar regulator loop (made of Ψ_R) which must be multiplied “by hand” by -1 compared to the ψ loop. If the axial-vector current is regularized as

$$j_\mu^{5,R} = \bar{\psi} \gamma_\mu \gamma_5 \psi - \bar{\Psi}_R \gamma_\mu \gamma_5 \Psi_R, \quad (243)$$

the classical equations of motion are applicable for $j_\mu^{5,R}$ provided that the equations of motion are derived from the total Lagrangian $\mathcal{L}_{\text{total}}$. Calculate $\partial^\mu j_\mu^{5,R}$, integrate out Ψ_R and find $\partial^\mu j_\mu^{5,R}$ in term of the photon field. Please, do not forget, that because of the gauge invariance A_μ can appear only through the field strength $F_{\mu\nu}$.

Answer

First, we write down the equations of motion for ψ , Ψ_R , and their conjugates:

$$i \not{D} \psi = 0, \quad (244)$$

$$(i \not{D} - M) \Psi_R = 0, \quad (245)$$

$$\bar{\psi} i \overleftarrow{\not{D}} = 0, \quad (246)$$

$$\bar{\Psi}_R (i \overleftarrow{\not{D}} + M) = 0. \quad (247)$$

Using these equations of motion, we can calculate the divergence of the original axial-vector current j_μ^5 :

$$\partial^\mu j_\mu^5 = \partial^\mu (\bar{\psi} \gamma_\mu \gamma_5 \psi) \quad (248)$$

$$= (\partial^\mu \bar{\psi}) \gamma_\mu \gamma_5 \psi + \bar{\psi} \gamma_\mu \gamma_5 (\partial^\mu \psi) \quad (249)$$

$$= \frac{1}{i} (-\bar{\psi} i \overleftarrow{\not{D}} \gamma_5 \psi + \bar{\psi} \gamma_5 i \not{D} \psi) \quad (250)$$

$$= 0. \quad (251)$$

Note that $\partial^\mu (\bar{\psi} \gamma_\mu \gamma_5 \psi) = \bar{\psi} \overleftarrow{D}^\mu \gamma_\mu \gamma_5 \psi + \bar{\psi} \gamma_\mu \gamma_5 D^\mu \psi$ because the terms proportional to the gauge field A_μ cancel out due to the opposite gauge charges of ψ and $\bar{\psi}$. Thus, classically, the axial-vector current is conserved.

Next, we calculate the divergence of the regularized axial-vector current $j_\mu^{5,R}$:

$$\partial^\mu j_\mu^{5,R} = \partial^\mu (\bar{\psi} \gamma_\mu \gamma_5 \psi - \bar{\Psi}_R \gamma_\mu \gamma_5 \Psi_R) \quad (252)$$

$$= \partial^\mu (\bar{\psi} \gamma_\mu \gamma_5 \psi) - \partial^\mu (\bar{\Psi}_R \gamma_\mu \gamma_5 \Psi_R) \quad (253)$$

$$= 0 - \frac{1}{i} (-\bar{\Psi}_R (i \overleftarrow{\not{D}} + M) \gamma_5 \Psi_R + \bar{\Psi}_R \gamma_5 (i \not{D} - M) \Psi_R) \quad (254)$$

$$= -2iM \bar{\Psi}_R \gamma_5 \Psi_R. \quad (255)$$

The "integration out" of Ψ_R means evaluating the expectation value of the operator obtained from the EOM:

$$\langle \partial^\mu j_\mu^{5,R} \rangle = -2iM \langle \bar{\Psi}_R \gamma_5 \Psi_R \rangle_A \quad (256)$$

This expectation value can be expressed using the fermion propagator in the gauge background $\mathcal{G}_A = (i \not{D} - M)^{-1}$:

$$\langle \bar{\Psi}_R(x) \gamma_5 \Psi_R(x) \rangle = -i \text{Tr} \left[\gamma_5 \langle x | \frac{1}{i \not{D} - M} | x \rangle \right]. \quad (257)$$

To evaluate the trace, we first rationalize the denominator of the propagator:

$$\frac{1}{i \not{D} - M} = \frac{i \not{D} + M}{(i \not{D})^2 - M^2} = \frac{i \not{D} + M}{-\not{D}^2 - M^2}. \quad (258)$$

Using the identity for the square of the Dirac operator in a gauge background:

$$\not{D}^2 = D^2 + \frac{i}{2} \sigma^{\mu\nu} F_{\mu\nu}, \quad (259)$$

where in two dimensions, $\sigma^{\mu\nu} = \frac{i}{2} [\gamma^\mu, \gamma^\nu] = i \epsilon^{\mu\nu} \gamma_5$. Thus, we have:

$$\not{D}^2 = D^2 - \frac{1}{2} \epsilon^{\mu\nu} F_{\mu\nu} \gamma_5. \quad (260)$$

Substituting this into the expression for the expectation value:

$$\langle \bar{\Psi} R \gamma_5 \Psi_R \rangle = -i \text{Tr} \left[\gamma_5 \frac{i \not{D} + M}{-D^2 + \frac{1}{2} \epsilon^{\mu\nu} F_{\mu\nu} \gamma_5 - M^2} \right]. \quad (261)$$

Since the trace over spinor indices $\text{tr}(\gamma_5 \gamma_\mu) = 0$, the $i \not{D}$ term in the numerator vanishes. We are left with:

$$\langle \bar{\Psi} R \gamma_5 \Psi_R \rangle = -i M \text{Tr} \left[\frac{\gamma_5}{-D^2 - M^2 + \frac{1}{2} \epsilon^{\mu\nu} F_{\mu\nu} \gamma_5} \right]. \quad (262)$$

Now, we expand the denominator for large M . Noting that we need a term proportional to γ_5 to get a non-vanishing spinor trace ($\text{tr}(\gamma_5^2) = \text{tr}(1) = 2$ in 2D, and $\text{tr}(\gamma_5) = 0$):

$$\frac{1}{(-D^2 - M^2) + \frac{1}{2} \epsilon^{\alpha\beta} F_{\alpha\beta} \gamma_5} \approx \frac{1}{-D^2 - M^2} - \frac{1}{(-D^2 - M^2)} \left(\frac{1}{2} \epsilon^{\alpha\beta} F_{\alpha\beta} \gamma_5 \right) \frac{1}{(-D^2 - M^2)} + \dots \quad (263)$$

Inserting the leading contributing term back into the trace:

$$\langle \bar{\Psi} R \gamma_5 \Psi_R \rangle \approx -i M \int \frac{d^2 k}{(2\pi)^2} \text{tr} \left[\gamma_5 \left(-\frac{\frac{1}{2} \epsilon^{\mu\nu} F_{\mu\nu} \gamma_5}{(-k^2 - M^2)^2} \right) \right] \quad (264)$$

$$= \frac{i M}{2} \epsilon^{\mu\nu} F_{\mu\nu} \cdot \text{tr}(1) \int \frac{d^2 k}{(2\pi)^2} \frac{1}{(k^2 + M^2)^2}. \quad (265)$$

The momentum integral in two dimensions is:

$$\int \frac{d^2 k}{(2\pi)^2} \frac{1}{(k^2 + M^2)^2} = \frac{1}{4\pi} \int_0^\infty \frac{du}{(u + M^2)^2} = \frac{1}{4\pi M^2}. \quad (266)$$

Combining everything, the expectation value becomes:

$$\langle \bar{\Psi} R \gamma_5 \Psi_R \rangle = \frac{i M \epsilon^{\mu\nu} F_{\mu\nu}}{4\pi M^2} = \frac{i \epsilon^{\mu\nu} F_{\mu\nu}}{4\pi M}. \quad (267)$$

Finally, we substitute this back into the expression for the divergence of the regularized current:

$$\langle \partial^\mu j_\mu^{5,R} \rangle = -2i M \langle \bar{\Psi} R \gamma_5 \Psi_R \rangle \quad (268)$$

$$= -2i M \left(\frac{i \epsilon^{\mu\nu} F_{\mu\nu}}{4\pi M} \right) \quad (269)$$

$$= \frac{1}{2\pi} \epsilon^{\mu\nu} F_{\mu\nu}. \quad (270)$$

Homework 12 Due to April 23 8:00 AM

Question 1

The standard model is formulated in terms of $4 + 8$ gauge bosons. The latter are gluons, the gauge bosons of QCD. Four gauge bosons represent the electroweak sector. Usually one starts with three gauge bosons W representing $SU(2)$, and one gauge boson B representing the hypercharge. Upon the spontaneous breaking W_3 and B mix with each other; their linear combinations form the Z boson and the massless photon γ . Derive the expressions for Z and γ in terms of W_3 and γ .

Basing on the anomalies' cancellations verified in Sec. 19.2 of my textbook, check that the cancelation persists for the triangle graphs for

$$ZW^+W^-, \quad ZZZ, \quad \gamma\gamma\gamma, \quad \gamma W^+W^-. \quad (271)$$

i.e. derive the analogs of Eqs. (19.3) – (19.10).

Answer

First, we need to express the Z boson and the photon γ in terms of the original gauge bosons W_3 and B . The mixing can be described by the Weinberg angle θ_W , which is defined as:

$$\tan \theta_W = \frac{g'}{g}, \quad (272)$$

where g is the $SU(2)$ gauge coupling and g' is the $U(1)$ gauge coupling. The physical gauge bosons are then given by:

$$Z_\mu = \cos \theta_W W_{3\mu} - \sin \theta_W B_\mu, \quad (273)$$

$$A_\mu = \sin \theta_W W_{3\mu} + \cos \theta_W B_\mu. \quad (274)$$

We will consider the amplitudes contribution from left-handed fermions and right-handed fermions separately. The sum of these contributions should vanish for the anomalies to cancel. The relevant Feynman diagrams involve triangle loops with fermions running in the loop and gauge bosons as external legs. Besides, because of the γ_5 matrix, their contributions from left-handed and right-handed fermions will have opposite signs.

- ZW^+W^- : Since $W^\pm$ bosons couple only to left-handed fermions ($g_R = 0$), only the left-handed doublets contribute to this triangle diagram. The coupling of the Z boson to left-handed fermions is $g_L^f = T_3^f - Q_f \sin^2 \theta_W$. The anomaly coefficient is proportional to the sum over all left-handed fermions in a

single generation:

$$\mathcal{A}_{ZW^+W^-} \propto \sum_{f \in L} g_L^f = \sum_{f \in L} T_3^f - \sin^2 \theta_W \sum_{f \in L} Q_f. \quad (275)$$

For one generation, the sums vanish independently:

$$\begin{aligned} - \sum_{f \in L} T_3^f &= (\tfrac{1}{2} - \tfrac{1}{2})_{lep} + 3 \times (\tfrac{1}{2} - \tfrac{1}{2})_{quark} = 0 \\ - \sum_{f \in L} Q_f &= (0 - 1)_{lep} + 3 \times (\tfrac{2}{3} - \tfrac{1}{3})_{quark} = -1 + 1 = 0 \end{aligned}$$

Thus, $\mathcal{A}_{ZW^+W^-} = 0$.

- ZZZ : The Z boson couples to both left-handed (g_L^f) and right-handed ($g_R^f = -Q_f \sin^2 \theta_W$) fermions. The total anomaly is proportional to $\sum_f [(g_L^f)^3 - (g_R^f)^3]$:

$$\mathcal{A}_{ZZZ} \propto \sum_f \left[(T_3^f - Q_f \sin^2 \theta_W)^3 - (-Q_f \sin^2 \theta_W)^3 \right]. \quad (276)$$

Expanding the cubic term $(T_3^f - Q_f \sin^2 \theta_W)^3$, we obtain:

$$\mathcal{A}_{ZZZ} \propto \sum_f \left[(T_3^f)^3 - 3(T_3^f)^2 Q_f \sin^2 \theta_W + 3T_3^f (Q_f \sin^2 \theta_W)^2 \right]. \quad (277)$$

Using $T_3 = \pm 1/2$ for doublets, each term vanishes upon summation over a generation:

$$\begin{aligned} - \sum T_3^3 &= \tfrac{1}{4} \sum T_3 = 0 \text{ (due to doublet structure).} \\ - \sum T_3^2 Q &= \tfrac{1}{4} \sum Q = 0 \text{ (due to lepton-quark charge cancellation).} \\ - \sum T_3 Q^2 &= \tfrac{1}{2} (Q_u^2 - Q_d^2) \times 3 + \tfrac{1}{2} (Q_\nu^2 - Q_e^2) = \tfrac{1}{2} (\tfrac{4}{9} - \tfrac{1}{9}) \times 3 + \tfrac{1}{2} (0 - 1) = \tfrac{1}{2} - \tfrac{1}{2} = 0. \end{aligned}$$

Consequently, $\mathcal{A}_{ZZZ} = 0$.

- $\gamma W^+ W^-$: Similar to $ZW^+ W^-$, only left-handed fermions contribute because of the W vertices. The photon couples to the charge Q_f regardless of chirality. The anomaly is:

$$\mathcal{A}_{\gamma W^+ W^-} \propto \sum_{f \in L} Q_f = (-1)_{lep} + 3 \times \left(\tfrac{1}{3} \right)_{quark} = 0. \quad (278)$$

This precise cancellation between leptons and quarks is essential for the consistency of the Standard Model.

- $\gamma\gamma\gamma$: The photon is a pure vector gauge boson, where $Q_L = Q_R = Q_f$. The anomaly contribution for each fermion species vanishes identically:

$$\mathcal{A}_{\gamma\gamma\gamma} \propto \sum_f (Q_f^3 - Q_f^3) = 0. \quad (279)$$

In conclusion, the anomaly cancellation persists for all the specified triangle graphs, ensuring the consistency of the Standard Model. $\square$